DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (A)
Osmania University, Hyderabad-07

CLASS TEST-II

Subject : Automata Languages & Computation

Class: B.E (CSE)
Semester: IV
Academic Year: 2024-2025

Max Marks: 30
Duration: 1 hr (12:00 noon to 1:30 p.m.)
Date : 25-06-2025

Sl. No.	PART A Answer All Questions (5 x 2 = 10 Marks)	Marks	BT	CO
1.	Use the grammar $S \rightarrow A1B$, $A \rightarrow 0A/\varepsilon$, $B \rightarrow 0B / 1B / \varepsilon$ to give leftmost and rightmost derivations for the string 00101	2	3	2
2.	Consider the CNF grammar G $S \rightarrow AB / BC$ $A \rightarrow BA / a$ $B \rightarrow CC / b$ $C \rightarrow AB / a$ Check the string ababa in L(G).	2	3	3
3.	Consider the grammar $S \rightarrow aS / aSbS / \varepsilon$ and show that whether it is ambiguous or not	2	3	2
4.	Give the construct of PDA P from empty stack to final state automaton.	2	2	2
5.	Consider the grammar $S \rightarrow AB / a$ $A \rightarrow b$. Eliminate useless symbols from this grammar.	2	3	2
	PART B **Answer Any Two Questions (2 x 10 = 20 Marks)**			
6.	Convert the grammar $S \rightarrow AAA / B$, $A \rightarrow aA / B$ $B \rightarrow \varepsilon$. Eliminate useless symbols if any, Eliminate ε – productions, and Eliminate unit productions put it into CNF form.	10	2	3
7.	Convert the grammar $S \rightarrow aAA$ $A \rightarrow aS/bS/a$ to a PDA that accepts the same language by empty stack.	10	3	3
8.	Design TM to implement multiplication by using subroutine.	10	3	4